UNIVERSITY OF FREIBURG

Department of Computer Science

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Machine Learning Course
Summer Semester 2022
Retake Exam

Parts	**Points**
1) Linear Methods and SVM	/ 10
2) Bias-variance tradeoff	/ 10
3) Decision Trees and Gradient-Boosted Decision Trees	/ 10
4) Neural Networks	/ 10
5) Ensembles/Boosting	/ 10
6) HPO and Error Measures	/ 10
Total	/ 60

This exam consists of 6 parts worth a total of **60 points** and you have **120 minutes** to answer the questions.

Please check the back of each page, the exam is printed on both sides.

This examination is closed book: **you are not allowed to use any notes**. Please put your name and matriculation number on every page.

Your name: _____ Table number: _____

Your matriculation number: _____

Your signature: _____

Good luck!

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Part 1 – Linear Methods and SVM (10P)

In the toy example of Fig. 1, linearly separable training data points are given.

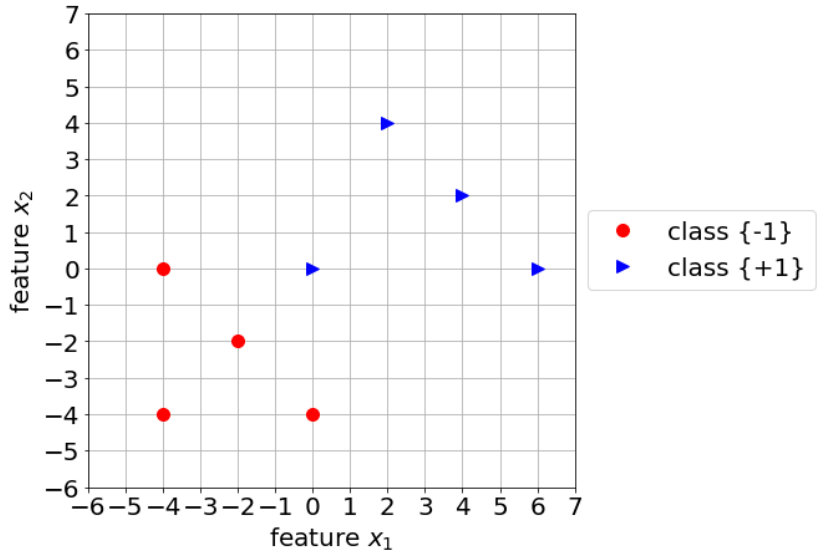


Figure 1: Toy data set #1.

a. **(2P)** Draw the optimal hyperplane to Fig. 1, which will be identified by a linear SVM. Write down the explicit function $f(x_1, x_2) = 0$ of this hyperplane based on the features x_1 and x_2. *Hint:* There is no need for any calculation.

b. **(6P)**

- **(3P)** Given that $\|\mathbf{w}\| = 1$ (L2-norm), calculate the values of the weight vector $\mathbf{w}$ and the bias b for the given data in Fig. 1 that the linear SVM optimization will deliver. Briefly describe your approach. What is the dimensionality of $\mathbf{w}$?

- **(1P)** Highlight all possible support vectors for the optimal linear SVM solution in Fig. 1. You can either mark them in the figure or write down their coordinates.

- **(2P)** Let's assume you know the parameters $\mathbf{w}$ and b of the primal form of a linear SVM. How can you assign the class for an unseen test data point x_{test}? Explicitly give a formula that calculates the label for x_{test}.

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c. **(2P)**

(a) **(1P)** Is vanilla SVM decision hyperplane sensitive to outliers in a linearly separable dataset? Briefly explain your answer.

(b) **(1P)** What would happen if we use vanilla SVM in a dataset that is not linearly separable?

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Part 2 – Bias-Variance Tradeoff and Regularization (10P)

a. **(2P)**

- **(1P)** Write (low or high) in the empty spaces below, for the connection between overfitting/underfitting and the Bias-Variance Tradeoff.

  Overfitting: _____ bias, _____ variance

  Underfitting: _____ bias, _____ variance

- **(1P)** Mention briefly two measures (half sentences) to **mitigate overfitting** in Neural Networks:

b. **(4P)** Assume that we trained 3 different models on a dataset with three samples $\mathcal{D} = \{x_1, x_2, x_3\}$ using the Mean Absolute Error (MAE) as loss function, and we test it on a sample x_*. The following table shows the prediction of the three trained models $(f^{(1)}, f^{(2)}, f^{(3)})$ on all the samples.

x	$f^{(1)}$	$f^{(2)}$	$f^{(3)}$	y
x_1	0.1	0.2	0	0.3
x_2	3.9	3.5	1	3.67
x_3	-0.9	-0.6	2	-0.7
x_*	2.0	0.1	0.7	2.3

According to the data in the table above:

- **(1P)** Compute the training error and the test error (MAE) for every model (i.e. for each $f^{(1)}, f^{(2)}, f^{(3)}$).

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- Answer the following empty spaces with one of the following options: $f^{(1)}, f^{(2)}$, or $f^{(3)}$. In addition, motivate your answers.

  - **(1P)** Which model exhibits **better generalization**?: _____

    Motivate:

  - **(1P)** Which model exhibits **underfitting**?: _____

    Motivate:

  - **(1P)** Which model exhibits **overfitting**?: _____

    Motivate:

c. **(4P)** Weight decay is a common approach for regularizing neural networks. Given that $\mathcal{L}, \theta, \mu$ are the loss function, network parameters, and learning rate respectively, it applies a gradient update as follows:

$$\theta_{t+1} = \theta_t - \alpha\theta_t - \mu\frac{\partial\mathcal{L}}{\partial\theta_t}$$

Show that the weight decay formula above is equivalent to the gradient descent update rule with the L2 regularization if $\alpha = 2\mu\lambda$, where λ is the L2 penalty term.

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Part 3 – Decision Trees and Gradient-Boosted Decision Trees (10P)

a. **(2P)**

(a) **(1P)**

- The **prediction function** of a decision tree with parameters $\theta =< R_1, ..., R_J, \hat{y}^{(R_1)}, ..., \hat{y}^{(R_J)} >$ is $f(x, \theta) = \sum_{j=1}^{J} \hat{y}^{(R_j)} \mathbb{I}(x \in R_j)$. What does R_J represent?
- Write the formula of Information Gain.

(b) **(1P)** Are Decision Trees sensitive to feature scaling? Explain.

b. **(4P)** Consider the dataset below with three features and a binary target variable (Default):

Home Owner	**Marital Status**	**Income** $<$ 80**K**	**Default**
Yes	Single	No	No
No	Married	No	No
No	Single	Yes	No
Yes	Married	No	No
No	Single	No	Yes
No	Married	Yes	No
No	Married	Yes	No
No	Single	No	Yes

- **(3P)** Using the CART algorithm, calculate the Information Gain for all the possible splits for a decision stump tree. Show the decision split with the **largest** Information Gain.

  Provided for ease of calculation:
  $log_2 \frac{3}{4} = -0.415 \quad log_2 \frac{1}{4} = -2 \quad log_2 \frac{2}{3} = -0.59 \quad log_2 \frac{1}{3} = -1.59 \quad log_2 \frac{1}{2} = -1 \quad log_2 \frac{2}{5} = -1.32 \quad log_2 \frac{3}{5} = -0.74$

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- **(1P)** What is the prediction of the decision stump for the test instance below?:

Home Owner	**Marital Status**	**Income** $<$ 80**K**	**Default**
No	Single	No	?

c. **(4P)** Mention one hyperparameter of Decision Trees that requires tuning. Describe the **bias and variance impact** of setting that hyperparameter to a (i) very **low**, or (ii) a very **high** value.

Hyperparameter:

(i) (2P) Impact of a very **low** hyperparameter value:

(ii) (2P) Impact of a very **high** hyperparameter value:

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Part 4 – Neural Networks (10P)

a. **(4P)**

- **(1P)** Mention one advantage and one disadvantage of using a small learning rate in Stochastic Gradient Descent.

- **(3P)** I would like to train neural networks for diagnosing the progressive severity of a patient's disease as "stage 1" (least advanced), "stage 2", "stage 3", and "stage 4" (most advanced).
  - (1P) How many neurons should I use at the last layer?
  - (1P) What activation function should I use at the last layer?
  - (1P) What loss function should I use to train the network?

b. **(3P)** You are given a dataset of 120 instances with 6 feature variables and one target variable, where the target has a domain of values {'Rainy', 'Sunny', 'Windy', 'Foggy'}. We would like to train a feed-forward neural network having 3 hidden layers with 16 units per layer, plus one output layer. No bias terms are used. What is the total number of parameters in this network?

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c. **(3P)** Consider training a neural network for a regression task, given M-dimensional input feature vectors $x \in \mathbb{R}^M$ and a target variable $y \in \mathbb{R}$. A feedforward neural network has $L > 3$ layers **and skip connections** as:

$$
\begin{aligned}
h^{(0)} &= x \\
h^{(i)} &= \sigma\left({w^{(i)}}^T h^{(i-1)}\right) && \forall i \in \{1, 2, L\} \\
h^{(k)} &= \sigma\left({w^{(k)}}^T h^{(k-1)}\right) + h^{(k-2)} && \forall k \in \{3, \ldots, L-1\} \\
\hat{y} &= h^{(L)}
\end{aligned}
$$

The network has no bias terms and the dimensions of the weight matrices are $w^{(1)} \in \mathbb{R}^{M \times D}, w^{(\ell)} \in \mathbb{R}^{D \times D}, \forall \ell \in \{2, \ldots, L-1\}$, and $w^{(L)} \in \mathbb{R}^{D \times 1}$. In other words, the number of neurons in all the hidden layers is D. Notice that the activations $h^{(k)}$ at layers $3 \leq k < L$ are inputted the activations of the two preceding layer $h^{(k-1)}$ and $h^{(k-2)}$ (the latter is called a skip connection). Additionally, the sigmoid function is $\sigma(x) = \frac{1}{1+e^{-x}}$ and the loss is the least-squares error $J(\hat{y}, y) = (\hat{y} - y)^2$.

What is the formula of the partial derivative $\frac{\partial J}{\partial h^{(k)}} = ?$ for a layer $3 \leq k < L$.

For simplicity assume all the terms $\frac{\partial J}{\partial h^{(k+1)}}, \ldots, \frac{\partial J}{\partial h^{(L)}}$ have been computed by the backpropagation algorithm and you can use them directly in the formula of $\frac{\partial J}{\partial h^{(k)}}$.

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Part 5 – Ensembles Boosting (10P)

a. **(2P)**

- **(1P)** Give one advantage and one disadvantage of using Random Forests instead of Gradient Boosted Decision Trees.

- **(1P)** Which of the following statements are true about Gradient Boosting? Write (**True** or **False**) on the left of the sentence.

  (a) Gradient Boosting uses multiple models to make predictions.

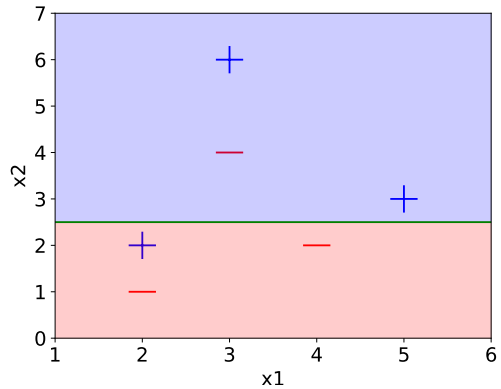
  (b) Gradient Boosting is insensitive to outliers.

  (c) Gradient Boosting is a sequential algorithm.

  (d) Training Gradient Boosted Decision Trees is less time-expensive than training a single Decision Tree.

b. **(4P) AdaBoost**. Consider the following dataset in the table at the left. A visualization of a decision stump tree (1 layer tree with one decision node) is shown in the figure next to the table. The decision split is done on $x_2 > 2.5$.

x_1	x_2	Class
2	1	minus
3	4	minus
3	6	plus
4	2	minus
2	2	plus
5	3	plus



Using the **AdaBoost** algorithm, calculate the **error** $Err_1 = \frac{\sum_{i=1}^{6} w_i^{(1)} \cdot \mathbb{I}(y \neq f^{(1)}(x_{i,1}, x_{i,2}))}{\sum_{i=1}^{6} w_i^{(1)}}$, **model weight** α_1 and updated **data point weights** $w_1^{(2)}, ..., w_6^{(2)}$. For the initial weights use $w_1^{(1)} = w_2^{(1)} = ... = w_6^{(1)} = \frac{1}{6}$.

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c. **(4P)** Consider the scenario of fitting an ensemble model on a classification dataset using Gradient-Boosted Decision Trees and Random Forests. Mention briefly (with one or two sentences) **one positive and one negative effect** of **increasing the number of models** for each of these algorithms.

- Gradient-Boosted Decision Trees:

  Positive:

  Negative:

- Random Forests:

  Positive:

  Negative:

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Part 6 – HPO and Error Measures (10P)

a. **(2P)** Why is it important to have an acquisition function that accounts for uncertainties?

b. **(4P)** Assume you are optimizing two hyperparameters of a feed-forward neural network (learning rate and the number of layers) for a dataset, and you have observed two configurations with the following evaluated validation accuracies:

Learning Rate	Number of Layers	Accuracy
0.1	3	0.8
0.5	1	0.4

You need to decide what configuration to evaluate next. In this exercise you have two choices among the following two configurations:

Learning Rate	Number of Layers	Accuracy
0.1	4	?
0.5	4	?

You also decide that you want to use Sequential Model-Based Optimization (SMBO) with a Random Forest surrogate. The Random Forest has two trees, each with only one decision stump that acts on only one of the hyperparameters, i.e. one tree uses the learning rate and the second tree uses the number of layers.

i. (1P) Write down the formula for computing the mean and the standard deviation of the estimated accuracy of a configuration when using a random forest with two trees.

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ii. (3P) Which configuration would you choose when using the Lower Confidence Bound (LCB) acquisition with $\beta = 1.1$ (Remember $LCB(x) = mean(x) + \beta \cdot std(x)$, and you want to maximize its value). Use the formula in part (a) to compute the mean and std.

c. **(4P)**

i. (1P) What is the exploration vs. exploitation trade-off in Hyperparameter Optimization?

ii. (1P) How did you trade off optimizing the hyperparameters of the networks in the previous exercise?

iii. (2P) Explain why a Random Forest having similar trees (with low diversity) will have a poor exploration.